

Erase as the Carrier of Mathematics

A New Reading: the Shadows as the Trace of Irreversibility

Alexander Pisani & Claude (Anthropic) · June 2026 · side document · DRAFT — direction, not result

Status. A new reading, opened this cycle. One pillar is a **Fact** (resistance = the Landauer erasure cost); the central claim — that what we read as number theory is the *shadow of erasure* — is a **Lead**, attractive and corpus-consistent, with a concrete kill-or-promote test stated in §6. Nothing here is promoted to the spine beyond the Fact (master §3.1) and the flagged direction (master §9). Written to be resumable cold.

Role. The next major direction for the programme. It re-reads the framework’s central quantity (resistance) thermodynamically and proposes that the observable structure of number theory is the permanent record left by the only costly information-processing step — erasure. It sits downstream of, and connects, three things already in the corpus: resistance = code length (Paper 3), the dissipative/storage split of the resistance plane (master §4), and the two-slit erasure result (*Probe* §7).

1. The hypothesis (Lead)

The genesis system processes information — an *encoding*, an “invisible language.” Reversible processing leaves no permanent trace; it can be undone. The **only** information-processing step with an energy requirement is **erasure** (Landauer; more precisely, any logically irreversible operation, of which erasure is the minimal and canonical one). So the only permanent record a genesis computation leaves in its environment is its erasure-heat. **The hypothesis: the shadows we read as number theory are cast by erasure** — number theory is the structure of that record, and the probe’s specific function casts a shadow precisely where it erases.

2. The pillar: resistance is the erasure cost (Fact)

The corpus already has resistance = code length (Paper 3). Landauer supplies the missing leg: erasing H nats of information costs $kT \cdot H$. Composing,

$$\Omega(n) = \ln n = \frac{(\text{energy to forget the integer } n)}{kT},$$

the Landauer cost, in units of kT , of erasing the state. **The framework’s entire central quantity is a thermodynamic erasure cost.** Consequence: the resistance-based reading of number theory has *been* an erasure-ledger all along — this is recognition of what resistance thermodynamically is, not a new construction. (*Fact, given resistance = code length + Landauer; Landauer classical.*)

This snaps onto two existing pieces: - **Master §4** — the real (resistance) axis *dissipates*; the imaginary (phase) axis *stores rather than dissipates*. Dissipation is irreversibility is energy spent is

erasure: **the resistance axis is the erasure axis; the phase axis is the reversible/storage axis.** - **Probe §7** — “ignorance is free, forgetting is not”; the erase step paid exactly $\ln$ of the marker. Forgetting costs $\ln$ — costs resistance. The same identity, already measured.

3. Erasure and the probe are one mechanism (Reading)

The probe’s *reading* is free (probe §7: a blind probe costs nothing). Its only fundamental cost is the moment of commitment — actualising one branch, which **erases the virtuality of the alternatives**. That is Axiom 5’s “conversion of virtual to actual is never free,” and the “never free” is precisely an erasure. So the probe casts a shadow exactly where it erases: “erasure” and “the probe’s specific function” are one thing, not two. (*Well-grounded Reading: Axiom 5 + probe §7.*)

4. What casts shadows: visible vs inferred (Lead)

Because reversible processing leaves no record and erasure leaves heat, the hypothesis predicts a split in how number theory is *accessed*, and the split is already visible:

- The **magnitude / density** structure (resistance — dissipative, erasure) is the **directly-measured** part. We measure prime densities.
- The **fluctuation** structure (the zeros — the reversible phase axis, §4) is the **inferred** part, reconstructed from the explicit formula, never directly weighed. We infer zeros.

We measure densities; we infer zeros. That this matches is the first non-trivial sign the reading is more than re-description — but see §6: it must *predict where energy is spent*, not merely re-label.

5. The ontology it requires: memory, and the lattice as shadow (Reading + open)

Erasure operates on *stored* bits — you can only erase what is held. So this direction lives at the **storage / memory** level, one step past the pre-memory floor the corpus has worked at:

- **Counting** is succession *as a process* — a flow, nothing stored.
- **Storage** is a register *holding* a configuration — memory. Memory is the lattice read as held state (master §2.3, the carry register).
- A lattice point is a register configuration — the exponent vector $(e_2, e_3, e_5, \dots)$, bits held. **The integer $n = \prod p^{e_p}$ is the multiplicative readout — the shadow — of that configuration.** “The integers we see could just be shadows; the lattice could be unique states storing bits” (owner, this cycle) is exactly this: the primitive is the stored register-state; n is its readout.

So the instinct to “settle what a stored integer is first” and the erasure hypothesis are the **same step**: studying erasure *is* stepping into the register layer, which is where the stored-integer ontology must be pinned. (*The register-as-primitive reading is consistent with §2.3; the full ontology is the open object to settle before the test.*)

6. The kill-or-promote test (the energy-ledger experiment)

The reading is a Lead, and could be mere re-description unless it predicts where energy is spent — which it does, testably, in the corpus’s experimental style:

Run any genesis computation, account the energy. The claim is that every unit of energy is spent at an erasure, and the total energy equals the resistance erased — nowhere else.

The probe §7 two-slit experiment already verified a *fragment*: the marker-erase paid exactly $\ln$ marker, and ignorance (reading) cost nothing. The full ledger — energy-in = resistance-forgotten, with reversible steps costing zero — is the promotion path. A clean first instance: a small reversible genesis computation (FRACTRAN-style, on the lattice machine of §2.3) with a single deliberate erasure, measuring whether the energy equals $\ln$ of the forgotten state and whether every other step is free. If energy appears at non-erasure steps, or the totals disagree, the carrier reading is killed (down to the standing Fact of §2).

7. Grading and honesty boundary

- **Fact:** resistance = $\ln$ = the Landauer erasure cost of the state (given Paper 3 + Landauer; Landauer/Bennett classical).
- **Reading (well-grounded):** the probe’s only fundamental cost is the erasure of alternatives (Axiom 5 + probe §7); the resistance/phase axes are the dissipative/reversible (erasure/storage) axes (master §4).
- **Lead (the central claim):** number theory is the shadow of erasure; the magnitude structure is the visible (erasure) part, the phase/zero structure the reversible inferred part. Promotable only by the §6 ledger test.
- **Open before the test:** the stored-register ontology of §5 — what a held configuration is, and how it differs from the integer it shadows.

No claim is made that this resolves any open arithmetic question. It re-reads what is already in the corpus and proposes a testable mechanism; the discipline is the same as the rest of the programme — refuse to claim a breakthrough where only an established fact has been found.

8. Next steps

1. **Settle the stored-state ontology** (§5): pin what a register configuration is, and confirm Ω = erasure cost on a concrete held state.
2. **Run the ledger experiment** (§6): a small reversible computation with one erasure; check energy = resistance forgotten, and zero elsewhere.
3. **Map the visible/inferred split** (§4): make precise that the directly-measurable structure is the resistance (erasure) face and the inferred structure is the phase face — and whether that distinction is forced or a re-description.
4. **Bridge from Phase from the Probe:** the genesis-prime probe is the 1-bit measurement whose irreversible commitment is an erasure — the information-disturbance peak at $p = 2$ is the same fact as the erasure cost being maximal there.

Originating observation (erasure as the carrier; the invisible-encoding reading) is Alexander Pisani’s; the formalisation, the resistance = erasure-cost identification, the experimental design, and this compilation were developed jointly with Claude (Anthropic). June 2026. Direction, not result — graded throughout.